

2025 UT Invitational
October 25th, 2025
Designer Genes C Exam



Parameters

- Each participant may bring 1 note sheet per team, and a scientific calculator.
- National level rules will be tested.

Directions

- You will have **50 minutes** to complete this exam - do not begin until instructed to do so!
- All answers must be indicated on your answer document. **DO NOT WRITE ON THIS EXAM!**
- Point values are indicated for all questions & all 5 tiebreakers are indicated in test booklets. The number of tiebreakers correctly answered indicates your tie order.
- The maximum number of points you can score on this exam is 135 points.
- You may take apart the test to split the work, but make sure to staple it back together at the end of the event.

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(any questions after the conclusion of this event may be emailed to jinjeong@umich.edu)

Multiple Choice (50 questions)

Questions 1-4 refer to the following image.

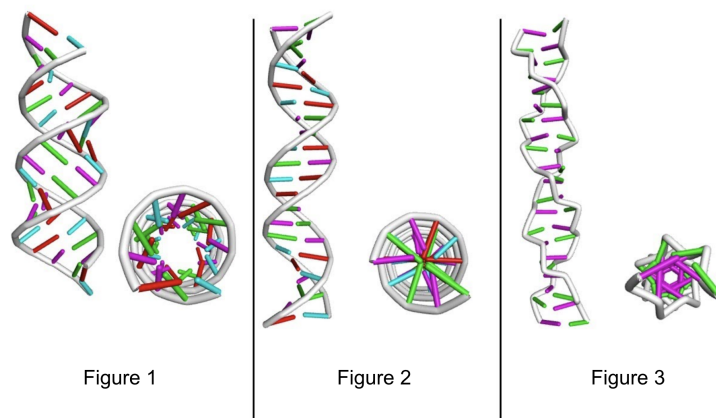


Image from:

<https://www.youtube.com/watch?v=9xhCbstatLA>

1. (1) Which figure shows the form of the DNA helix under normal physiological conditions?
 - A. Figure 1
 - B. Figure 2
 - C. Figure 3
 - D. None of the above
2. (1) Which of the following sequences might be an example of a segment of dsDNA that forms Figure 3? (Select all that apply.)
 - A. ACTCTGAC
 - B. ATATATAT
 - C. AACCTTCC
 - D. GCTAGCGC
 - E. None of the above
3. (1) Which figure(s) show right-handed helices? (Select all that apply.)
 - A. Figure 1
 - B. Figure 2
 - C. Figure 3
 - D. All of the above
 - E. None of the above
4. (1) **TIEBREAKER** Which figure shows a helix that can be an RNA-DNA hybrid?
 - A. Figure 1
 - B. Figure 2
 - C. Figure 3
 - D. None of the above

5. (1) The following template strands are exposed to primers, excess nucleotides, and DNA polymerase in buffer at approx. 60°C. The resulting strands of dsDNA are then incubated at increasing temperatures for 30 minutes. Which primer's corresponding dsDNA segment would likely require the *highest* denaturing temperature?
- A. AATACGCTA
 - B. GATGCTCTA
 - C. TTACTAGTA
 - D. CTACGCCAG
6. (1) Why does the dsDNA strand you chose in the previous question have the highest denaturing temperature?
- A. There is a high percentage of A-T bonds, which have less hydrogen bonds to break.
 - B. There is a high percentage of G-C bonds, which have more hydrogen bonds to break.
 - C. There is a high percentage of G-C bonds, which have more hydrogen bonds and π -stacking interactions to break.
 - D. There is a high percentage of A-T bonds, which have more hydrogen bonds and π -stacking interactions to break.
7. (1) What type of bond links together nucleotides in the phosphate backbones of DNA and RNA?
- A. Disulfide bridges
 - B. Hydrogen bonds
 - C. Phosphodiester bonds
 - D. Peptide bonds
8. (1) The bonds in the previous question form between:
- A. The phosphate group on the 3' carbon and the hydroxyl group on the 5' carbon
 - B. The phosphate group on the 5' carbon and the hydroxyl group on the 3' carbon
 - C. C' and the amino group
 - D. The bases of each nucleotide
 - E. The active site of DNA polymerase and the major groove
9. (1) Sanger sequencing relies on the incorporation of _____ into the DNA chain being elongated.
- A. dNTPs
 - B. ddNTPs
 - C. NTPs
 - D. AMP
10. (1) Which of the following does *not* correctly describe one of Mendel's Laws of Inheritance?
- A. During meiosis, homologous chromosome pairs line up at the equator of the cell. Each pair's place in the order is not influenced by any other pair's place in the order.
 - B. A brown-haired person with attached earlobes has children with a red-haired person who has free earlobes. Their children all have brown hair and dimples.
 - C. During meiosis, chromosome pairs remain together through metaphase. Pairs of alleles are separated during anaphase and end up in different gametes.
 - D. A professional arm wrestler develops huge arm muscles on his right, dominant arm. His children all have larger right arms at birth.

11. (1) Which of the following describes a *deviance* from Mendel's Laws of Inheritance?
- A. A flood kills most individuals of a species of birds living on a remote island. The survivors all have waterproof feathers and a tendency to nest on treetops.
 - B. A giraffe stretches its neck to reach increasingly high branches of leaves over the course of its life. Its offspring have slightly longer necks than their parents.
 - C. A right-handed parent and a left-handed parent have children. Three of their four children are right-handed.
 - D. A red flowering plant and a white flowering plant can be crossbred to produce pink flowered offspring.
12. (1) In a certain species of spiny crabs, the allele for spiked claws (*S*) is dominant over the allele for smooth claws (*s*). Expression of another gene, *Smh1*, prevents spiny crabs' claws from developing. The effect that *Smh1* has on claw spikiness is called:
- A. epigenetics
 - B. epistasis
 - C. epinephrine
 - D. hypostasis
13. (1) Cells are ___ at the beginning of mitosis, and ___ after cytokinesis.
- A. $2n$, $2n$
 - B. $4n$, n
 - C. $4n$, $2n$
 - D. $2n$, n
14. (1) **TIEBREAKER** Immunoglobulin class switching:
- A. affects antigen specificity.
 - B. changes the constant region of the antibody heavy chain.
 - C. does not affect interactions with effector molecules.
 - D. changes the variable region of the antibody heavy chain.
15. (1) Which of the following is not a correct assumption of the Hardy-Weinburg equilibrium?
- A. Free genetic flow
 - B. No mutations
 - C. Large population
 - D. Random mating
16. (1) Cells at the beginning of meiosis I are ____, and cells at the beginning of meiosis II are ____.
- A. diploid, haploid
 - B. $4n$, n
 - C. $2n$, n
 - D. $2n$, $2n$

17. (1) In a tropical storm, a small population of lemurs is swept out to a remote island. They reproduce and begin to populate the island. After a few generations, the island lemurs tend to be smaller on average than the mainland lemurs. This is an example of:
- A. Mutation
 - B. Gene flow
 - C. Bottleneck
 - D. Founder effect
18. (1) On the same island, another tropical storm drowns most of the island lemurs. The remaining lemurs repopulate the island. After a few generations, the lemur population now tends to have shorter fur that dries faster. This is an example of:
- A. Mutation
 - B. Gene flow
 - C. Bottleneck
 - D. Founder effect
19. (1) Scientists want to study a certain protein that is known to have a critical role in the cell cycle, and has recently been implicated in cancer. Overexpression of this protein is hypothesized to contribute to irregular cell cycle entry. What experimental technique is best suited to testing the hypothesis?
- A. Overexpression
 - B. Knockout
 - C. Knockdown
 - D. Knockin
20. (1) Which is not an appropriate situation to use RNA-seq?
- A. Measuring expression levels of multiple genes
 - B. Identifying localization of expressed genes
 - C. Detecting rare variants of a specific gene
 - D. Identifying coregulated genes
21. (1) A scientist suspects that a gene that encodes mouse ear shape (*Esh5*) is linked to a gene that controls fur length (*Fuzz1*). *Esh5* is known to be localized to chromosome 3q21. What technique should the scientist use in order to determine how close the locus of *Fuzz1* is to the locus of *Esh5* on chromosome 3?
- A. Hi-C
 - B. Minute Maid
 - C. scRNA-seq
 - D. ChIP-seq
22. (1) During bacterial transformation, what is the purpose of heat shock?
- A. To perforate the nuclear membrane of the bacteria.
 - B. To give the bacteria mutations that will be studied in the cloned organism.
 - C. To make the bacteria take up a recombinant plasmid.
 - D. To disinfect the bacteria and make it noninfectious.

23. (1) What is an accurate comparison between next-generation and third-generation sequencing?
- A. NGS has higher error rates than TGS.
 - B. NGS allows short reads, while TGS allows longer reads.
 - C. NGS is more expensive than TGS.
 - D. NGS is ideal for haplotype phasing, while TGS struggles with complex regions.
24. (1) At the end of PCR, amplified segments are usually refined using:
- A. Western blot
 - B. Size-exclusion chromatography
 - C. Gel electrophoresis
 - D. SDS-PAGE
25. (1) Which of the following is not true of additive alleles?
- A. They contribute to polygenic traits.
 - B. They can be affected by epistasis.
 - C. Each allele's contribution to the trait they encode is independent to the other alleles.
 - D. They contribute to a continuous range of variation.
26. (1) Two segments of DNA that are in two different species but share a common ancestor are:
- A. Homologs
 - B. Paralogs
 - C. Analogs
 - D. Orthologs

Exam continues on next page.

Questions 27-28 pertain to the below figure.

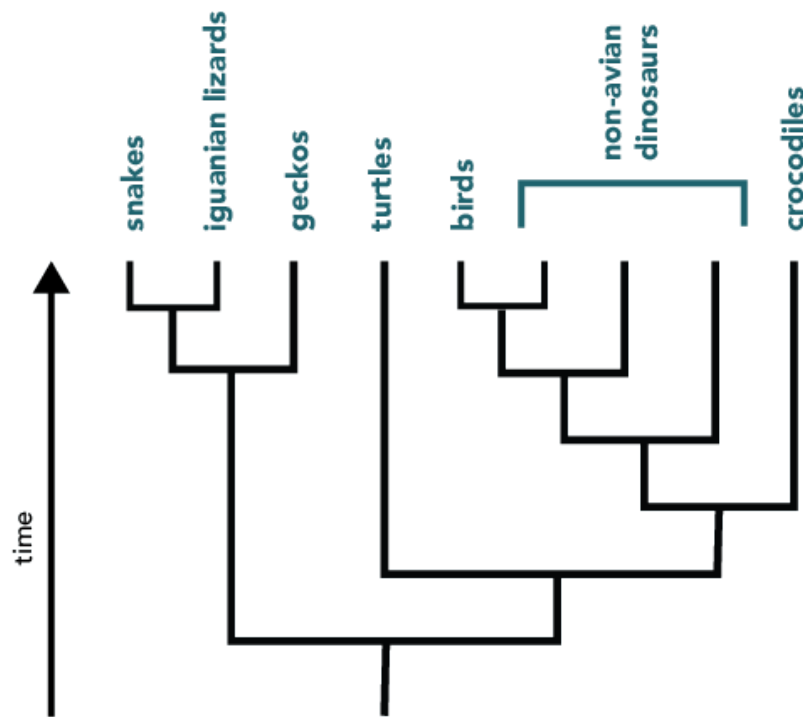


Image from:

<https://evolution.berkeley.edu/evolution-101/the-history-of-life-looking-at-the-patterns/using-the-tree-for-classification/>

27. (1) Which species have the most recent common ancestor?
- Snakes and iguanian lizards
 - Geckos and snakes
 - Crocodiles and non-avian dinosaurs
 - Turtles and crocodiles
28. (1) Which species likely has the least DNA in common with the others?
- Turtles
 - Crocodiles
 - Birds
 - Geckos
29. (1) A mutation that changes the protein expressed by a certain gene from Met-Ala-His-Pro-Ser to Met-Gly-Ile-His is most likely:
- nonsense
 - missense
 - frameshift
 - silent

30. (1) The mutation in the previous question was likely caused by:
- A. single base insertion
 - B. single base deletion
 - C. single base substitution
 - D. chromosomal rearrangement
31. (1) Which of the following correctly matches the class of RNA to its function? (Select all that apply.)
- A. tRNA: brings amino acids to ribosomes during translation
 - B. mRNA: transports genetic information to cytoplasm for protein synthesis
 - C. miRNA: regulates expression of mitochondrial DNA
 - D. rRNA: regulates gene expression in the nucleus
32. (1) During plasmid transformation, GFP and genes encoding antibiotic resistance are often added to the plasmid in order to:
- A. Select plasmids that were properly transformed.
 - B. Differentiate cloned organisms from control organisms.
 - C. Have fun :)
 - D. Have an epistatic effect on the gene being studied.
33. (1) Which of the following types of mutations is potentially the most deleterious?
- A. Missense in gene encoding protein integral to cell cycle regulation
 - B. Frameshift in gene encoding protein integral to cell cycle regulation
 - C. Nonsense in noncoding segment of DNA
 - D. Silent in gene encoding protein involved in embryonic development
34. (1) Which statement about DNA replication is not true?
- A. The leading strand is synthesized 5' to 3' towards the replication fork.
 - B. Replication occurs during the S phase of the cell cycle, and results in a diploid cell.
 - C. DNA polymerase I synthesizes most of the new DNA.
 - D. Helicase and topoisomerase make the replication fork accessible to the involved proteins.
35. (1) A missense mutation in topoisomerase approximately 500bp following its active site would likely result in:
- A. A nonfunctional topoisomerase
 - B. A functional topoisomerase with slightly altered folding
 - C. A truncated topoisomerase that is still functional
 - D. A normal, functional topoisomerase
36. (1) Mistakes made during DNA replication are corrected by:
- A. The endonuclease action of ligase
 - B. The endonuclease action of DNA pol III
 - C. The exonuclease action of DNA pol I
 - D. The exonuclease action of DNA pol III

37. (1) What is the correct order of DNA organization, from smallest to largest?
- A. Chromosome → Chromatid → Chromatin → Nucleosome → Double helix
 - B. Chromatid → Heterochromatin → Euchromatin → Chromosome → Double helix
 - C. Double helix → Nucleosome → Chromatin → Chromatid → Chromosome
 - D. Double helix → Heterochromatin → Euchromatin → Chromosome
38. (1) DNA damage caused by deamination would be repaired using:
- A. Non-homologous end joining
 - B. Homologous recombination
 - C. Base excision repair
 - D. Nucleotide excision repair
39. (1) What are two reasons why transcription is necessary to the central dogma? Why not simply synthesize proteins directly from DNA?
- A. Transcribing genetic information to RNA creates less errors.
 - B. Synthesized proteins cannot be transported out of the nucleus, where DNA lives.
 - C. Gene expression is regulated at the start of transcription.
 - D. We need to keep mRNA employed to help the economy.
40. (1) Which kind of RNA is not involved in the elongation step of translation?
- A. tRNA
 - B. mRNA
 - C. rRNA
 - D. miRNA
41. (1) In the presence of lactose, a *lac* operon:
- A. is activated by allolactose inactivating the repressor
 - B. is activated by allolactose activating the operator
 - C. is deactivated by allolactose activating the repressor
 - D. is deactivated by allolactose inactivating the operator
42. (1) **TIEBREAKER** Which of the following statements is incorrect?
- A. The Sec system secretes unfolded proteins, while the Tat system secretes folded proteins.
 - B. Proteins being exported through the Sec pathway are transported through SecYEG translocase.
 - C. The Sec system is primarily used by prokaryotes, while the Tat system is primarily used by eukaryotes.
 - D. Gram-negative bacteria can secrete proteins into the periplasm or extracellularly with the Tat system.
43. (1) Which of the following statements is true?
- A. Euchromatin is more dense than heterochromatin.
 - B. Heterochromatin is stabilized by DNA methylation.
 - C. Euchromatin has hypoacetylated histones.
 - D. Heterochromatin has acetylated histones.

44. (1) Which of the following situations does not violate the Hardy-Weinberg assumptions?
- A. A colony of bacteria in a lab are given excess resources so that they can reproduce indefinitely.
 - B. A herd of zebra migrate across a river; their new territory gives them access to another herd of zebra.
 - C. A royal family practices strict arranged marriages for their offspring to other noble families.
 - D. A farmer thins out their rows of corn so that only the tallest stalks can grow.
45. (1) In order to affect many target genes on different chromosomes, you should activate:
- A. A promoter
 - B. An enhancer
 - C. A silencer
 - D. A transcription factor
46. (1) **TIEBREAKER** Which of the following mechanisms can cause frameshift mutations? (Select all that apply.)
- A. Depurination
 - B. UV light damage
 - C. Ionizing radiation
 - D. Mutagenic chemicals
47. (1) A *trp* operon is inactivated when:
- A. a lack of tryptophan prevents the activator from being activated.
 - B. excess tryptophan binds to the repressor, causing it to block the operator site.
 - C. excess tryptophan inactivates the activator.
 - D. a lack of tryptophan allows the repressor to bind to and block the operator site.
48. (1) Is the following nitrogenous base found in RNA or DNA?

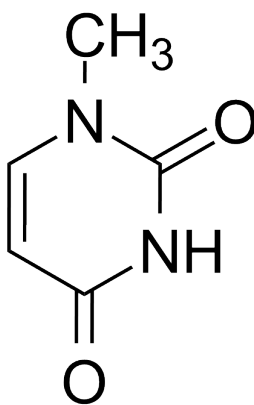


Image from: georganics.com

- A. DNA
- B. RNA
- C. Both
- D. Neither

49. (1) Sequence-specific protein-DNA interactions happen in the:
- A. Major groove
 - B. Minor groove
 - C. Phosphate backbone
 - D. All of the above
50. (1) Which of the following are wobble pairs? (Select all that apply.)
- A. G-U
 - B. A-C
 - C. T-C
 - D. A-G

Short Response (7 questions)

51. In mice, the agouti gene controls variations in color. There are 17 alleles of the agouti gene. One of the alleles, lethal yellow (A^y), is dominant over the nonagouti allele (a) and causes the mouse to have yellow fur and obesity. Homozygous lethal yellow is lethal early in development.
- (a) (8) A mature female mouse with lethal yellow coloring is crossed with a non-agouti (wildtype) male mouse. Fill in the Punnett square below to show the possible genotypes for the resulting offspring.

[illegible]

Image from: <https://www.kamiapp.com/library/resources/66k6uGj1GVby-punnett-square>

- (b) (3) If there are three pups in the litter, what is the probability that all three are heterozygous? Show all work.
- (c) (3) Lethal yellow offspring can be rescued in utero by feeding the dam a diet supplemented with methyl group-rich compounds. How is this possible?
- (d) (4) Suppose a scientist wants to study the process described in the previous question that allows for lethal yellow mice's offspring to display the wildtype phenotype. What would be the best way to precisely **knockout** a heterozygous offspring's A^y allele? Briefly explain why you chose this technique.

52. A certain breed of dog is prized for having curly fur. A mated pair and their offspring were observed for three generations, as shown in the following pedigree.

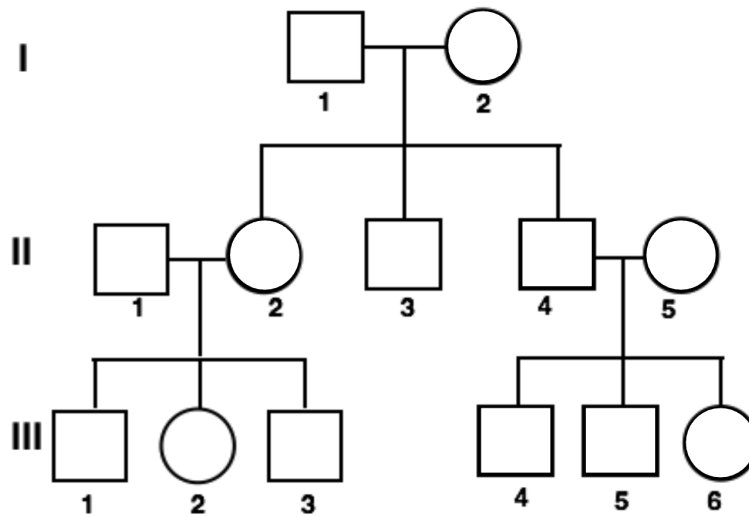


Image from: Khan Academy; Pedigrees

- (a) (4) Fill out the pedigree using the following information:
1. Parent I-2 has curly fur.
 2. Offspring II-2 has curly fur.
 3. Offspring II-3 has curly fur.
 4. Offspring III-5 has curly fur.
- (b) (6) Mark confirmed carriers with an X.
- (c) (2) What pattern of inheritance does the curly fur gene follow?

53. Subject 1 has a child, Subject 2, who has presented with Down Syndrome. Subject 1 and Subject 2's other parent both present as phenotypically normal. Below is a karyotype of a sample taken from Subject 1:

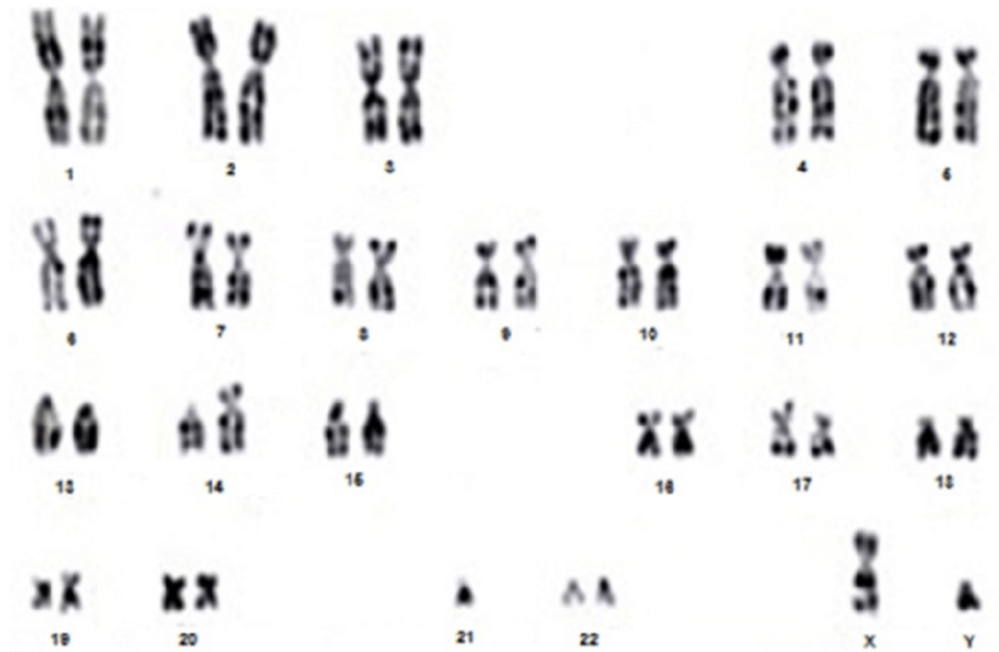


Image from: https://www.researchgate.net/publication/321366246_Cytogenetic_study_is_not_essential_in_patients_with_aplastic_anemia

- (a) (5) Based on the karyotype and the given background for Subject 1, determine the *most likely* genetic anomaly.
- (b) (1) True or false: All of this individual (Subject 1)'s cells *must* display the same chromosomal anomaly. A. True B. False
- (c) (5) What was the likely mechanism of inheritance of Down Syndrome in Subject 2?

54. Refer to the figure below.

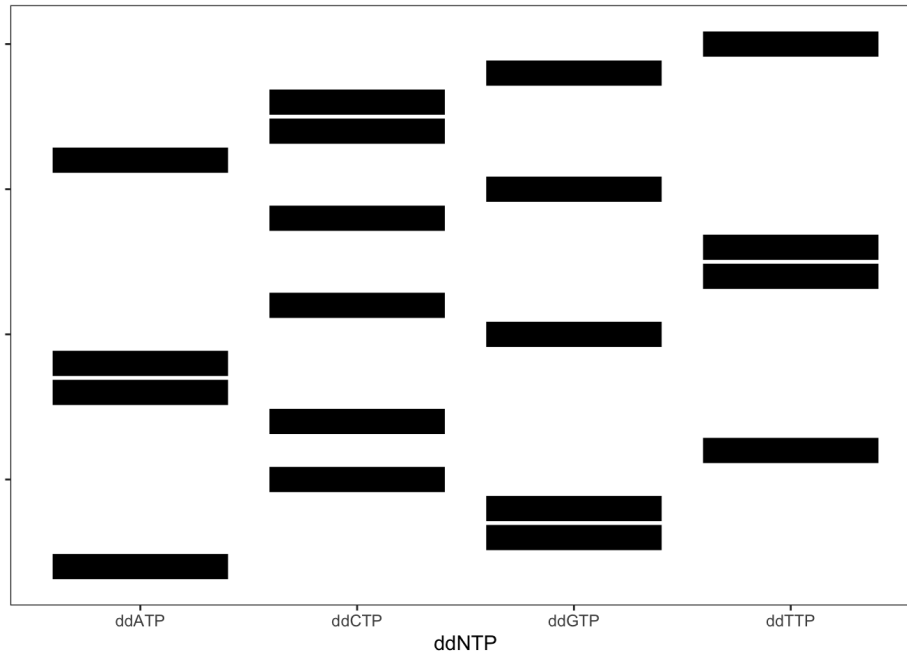


Image from: <https://eriqande.github.io/eca-bioinf-handbook/dna-sequences-and-sequencing.html>

- (a) (2) What type of laboratory technique produced the results shown in the figure?
- (b) (2) These readings represent the output of which method of DNA sequencing?
- (c) (5) Interpret the results in the figure. (Hint: you should get a 5' to 3' DNA sequence.)
- (d) (5) Convert the results to the corresponding mRNA strand that would be produced from transcription. (Write your answer 5' to 3'.)
55. Suppose you are interested in measuring recombination frequency between the following two genes in *Drosophila*:

snoot, with a dominant *st+* allele that encodes long proboscis and the recessive *st* that encodes truncated proboscis.

bzz2, with a dominant *bz+* allele that encodes normal wings and a recessive *bz* allele that encodes vestigial wings.

- (a) (6) Describe the two crosses you would need to perform in order to observe recombination events.
- (b) (5) Four phenotypes were observed in the offspring: long proboscis and normal wings, long proboscis and vestigial wings, short proboscis and normal wings, and short proboscis and vestigial wings. The following table shows the number of offspring with each phenotype:

Table 1: Frequency of phenotypes in recombinant *Drosophila* offspring

	<i>st+ bz+</i>	<i>st bz</i>	<i>st+ bz</i>	<i>st bz+</i>
Frequency	1339	1195	151	154

Calculate the recombination frequency between the *snoot* and *bzz2* genes. Show all work.

- (c) (3) **TIEBREAKER** Suppose we want to compare *snoot* and *bzz2* to *reb1*, a gene which specifies whether *Drosophila* eyes are red or black. Given the following recombination frequencies and the RF between *snoot* and *bzz2* you calculated in part b), determine the order of the genes on the chromosome.

$$\text{RF}(\textit{snoot-reb1}) = 9.1\%$$

$$\text{RF}(\textit{bzz2-reb1}) = 19.8\%$$

56. Two inbred lines of peas were intercrossed; the variance in pod length among F1 was 2mm. The F1 offspring were intercrossed; the variance in pod length in F2 was 7mm.

- (a) (5) Calculate the broad-sense heritability of pod length in the F2 population of peas.
(b) (4) Interpret the h^2 you calculated in the previous part.

57. In a local population of sea lions, two coat patterns can be observed. *Cc* and *CC* animals have spotted coats, while *cc* animals have solid coats. The C allele has a frequency of 0.3, while the c allele has a frequency of 0.7. Predict the genotype frequencies in this population of sea lions.